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GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Probability and Statistics

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

46	25	2005 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	23 questions
Marks 2	23 questions
Answer key	at the end

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Marks 1 23 questions

Q1. GATE ECE 2025

A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement.

What is the probability that the two balls drawn have different colours?

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 1

Open solution

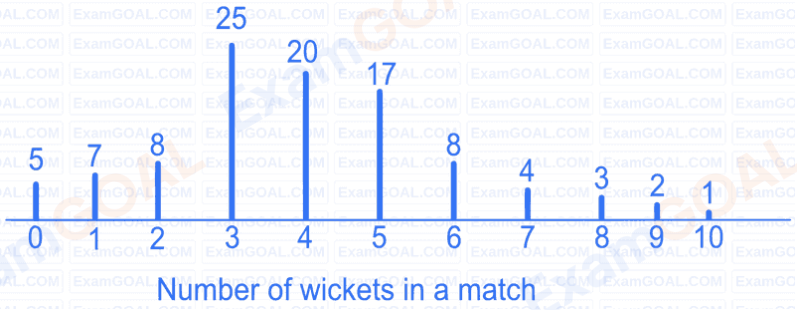
Q2. GATE ECE 2024

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval $[0, 1]$. The probability that $X \geq Y$ is _____ .

Open solution

Q3. GATE ECE 2022

The bar graph shown the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is _____ (rounded off to one decimal place).



Open solution

Q4. GATE ECE 2021

Two continuous random variables X and Y are related as Y = 2X + 3. Let σ_x^2 and σ_y^2 denote the variances of X and Y, respectively. The variances are related as

(A) $\sigma_Y^2 = 25\sigma_X^2$ (B) $\sigma_Y^2 = 4\sigma_X^2$

(C) $\sigma_Y^2 = 2\sigma_X^2$ (D) $\sigma_Y^2 = 5\sigma_X^2$

Open solution

Q5. GATE ECE 2020

The two sides of a fair coin are labelled as 0 and 1 . The coin is tossed two times independently. Let M and N denote the labels corresponding to the outcomes of those tosses. For a random variable X, defined as $X = \min(M, N)$, the expected value $E[X]$ (rounded off to two decimal places) is ____ .

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Q6. GATE ECE 2018

Let X_1 , X_2 , X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is _____.

Open solution

Q7. GATE ECE 2017 Set 2

500 students are taking one or more courses out of chemistry, physics and Mathematics. Registration records indicate course enrolment as follows: chemistry (329), physics (186), Mathematics (295), chemistry and physics (83), chemistry and Mathematics (217), and physic and Mathematics (63), How many students are taking all 3 subjects?

(A) 37 (B) 43 (C) 47 (D) 53

Open solution

Q8. GATE ECE 2017 Set 1

Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (up to third decimal place) _____.

Open solution

Q9. GATE ECE 2016 Set 1

The second moment of a Poisson-distributed random variables is 2. The mean of the random variable is _____.

Open solution

Q10. GATE ECE 2015 Set 3

The variance of the random variable X with probability density function $f(x) = \frac{1}{2}|x|e^{-|x|}$ is _____.

Open solution

Q11. GATE ECE 2015 Set 2

Let the random variable X represent the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. The expectation of X is _____.

Open solution

Q12. GATE ECE 2015 Set 2

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $1/6$ and that of Ramesh is $1/8$. What is the probability that only one of them will be selected?

(A) $47/48$ (B) $1/4$ (C) $13/48$ (D) $35/48$

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Q13. GATE ECE 2015 Set 1

Suppose A & B are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$. Let $\overrightarrow{A}$ & $\overrightarrow{B}$ be their complements. Which of the following statements is FALSE?

(A) $P(A \cap B) = P(A)P(B)$

(B) $P(A/B) = P(A)$

(C) $P(A \cup B) = P(A) + P(B)$

(D) $P\left(\overrightarrow{A} \cap \overrightarrow{B}\right) = P\left(\overrightarrow{A}\right).P\left(\overrightarrow{B}\right)$

Open solution

Q14. GATE ECE 2014 Set 1

In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is _____.

Open solution

Q15. GATE ECE 2014 Set 1

Let X be a real - valued random variable with $E[X]$ and $E\left[X^2\right]$ denoting the mean values of X and X^2, respectively. The relation which always holds true is

(A) $(E[X])^2 > E\left[X^2\right]$

(B) $E\left[X^2\right] \geq (E[X])^2$

(C) $E\left[X^2\right] = (E[X])^2$

(D) $E\left[X^2\right] > (E[X])^2$

Open solution

Q16. GATE ECE 2014 Set 4

Let X be a zero mean unit variance Gaussian random variable. $E[|X|]$ is equal to _____

Open solution

Q17. GATE ECE 2014 Set 4

If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier of future calls, the probability distribution function of the total number of calls in a fixed time interval will be

(A) Poisson

(B) Gaussian

(C) Exponential

(D) Gamma

Open solution

Q18. GATE ECE 2014 Set 3

An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is

(A) 0.067

(B) 0.073

(C) 0.082

(D) 0.091

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Q19. GATE ECE 2012

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that max $[X, Y]$ is less than $1/2$ is

(A) $3/4$ (B) $9/16$ (C) $1/4$ (D) $2/3$

Open solution

Q20. GATE ECE 2011

A fair dice is tossed two times. The probability that the 2nd toss results in a value that is higher than the first toss is

(A) $\frac{2}{36}$ (B) $\frac{2}{6}$ (C) $\frac{5}{12}$ (D) $\frac{1}{2}$

Open solution

Q21. GATE ECE 2009

A fair coin is tossed 10 times. What is the probability that only the first two tosses will yield heads?

(A) $\left(\frac{1}{2}\right)^2$ (B) $10c_2 \left(\frac{1}{2}\right)^2$
(C) $\left(\frac{1}{2}\right)^{10}$ (D) $10c_2 \left(\frac{1}{2}\right)^{10}$

Open solution

Q22. GATE ECE 2007

If E denotes expectation, the variance of a random variable X is given by

(A) $E\left(X^2\right) - E^2(X)$ (B) $E\left(X^2\right) + E^2(X)$
(C) $E\left(X^2\right)$ (D) $E^2(X)$

Open solution

Q23. GATE ECE 2005

A fair dice is rolled twice. The probability that an odd number will follow an even number is

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Open solution

Marks 2 23 questions

Q24. GATE ECE 2026

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows Normal distribution with zero mean and unity variance.

Y and Z are defined as, $Y = X + N$ and $Z = X^2 + N$.

Which of the following pairs represents the values of correlation between X and Y and that between X and Z ?

(A) $1/3$ and 0 (B) $1/3$ and $1/9$ (C) $1/3$ and $1/3$ (D) 1 and 0

Open solution

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Q25. GATE ECE 2025

Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained. The expectation of X is _____ (rounded off to two decimal places).

Open solution

Q26. GATE ECE 2021

In a high school having equal number of boy students and girl students, 75% of the students study Science and the remaining 25% students study Commerce. Commerce students are two times more likely to be a boy than are Science students. The amount of information gained in knowing that a randomly selected girl student studies Commerce (rounded off to three decimal place) is _____ bits.

Open solution

Q27. GATE ECE 2021

A box contains the following three coins.

I. A fair coin head on one face and tail on the other face.

II. A coin with heads to both the faces.

III. A coin with tails on both the faces.

A coin is picked randomly from the box and tossed. Out of the two remaining coins in the box, one coin is then picked randomly and tossed. If the first toss results in a head, the probability of getting a head in the second toss is

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Open solution

Q28. GATE ECE 2020

X is a random variable with uniform probability density function in the interval $[-2, 10]$. For $Y = 2X - 6$, the conditional probability $P(Y \leq 7 \mid X \geq 5)$ (rounded off to three decimal places) is ____ .

Open solution

Q29. GATE ECE 2017 Set 2

Passengers try repeatedly to get a seat reservation in any train running between two stations until they are successful. If there is 40% chance of getting reservation in any attempt by a passenger, then the average number of attempts that passengers need to make to get a seat reserved is _____.

Open solution

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Q30. GATE ECE 2016 Set 1

Two random variables X and Y are distributed according to

$$f_{X,Y}(x,y) = \begin{cases} (x+y), & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

The probability $P(X+Y \le 1)$ is _____.

Open solution

Q31. GATE ECE 2016 Set 3

The probability of getting a "head" in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a "head" is obtained. If the tosses are independent, then the probability of getting "head" for the first time in the fifth toss is _____.

Open solution

Q32. GATE ECE 2015 Set 3

A fair die with faces $\{1,2,3,4,5,6\}$ is thrown repeatedly till $'3'$ is observed for the first time. Let X denote the number of times the dice is thrown. The expected value of X is _____.

Open solution

Q33. GATE ECE 2015 Set 2

Let $X \in \{0,1\}$ and $Y \in \{0,1\}$ be two independent binary random variables. If $P(X=0) = p$ and $P(Y=0) = q$, then $P(X+Y \ge 1)$ is equal to

(A) $pq + (1-p)(1-q)$

(B) pq

(C) $p(1-q)$

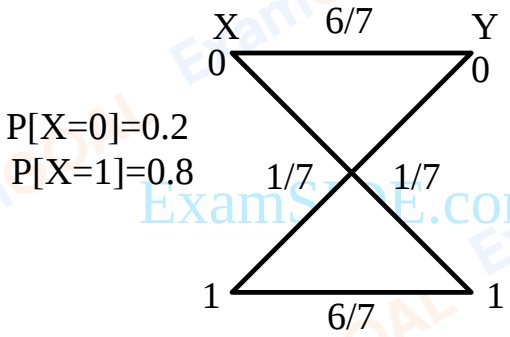
(D) $1-pq$

Open solution

Q34. GATE ECE 2015 Set 1

The input X to the Binary Symmetric Channel (BSC) shown in the figure is $'1'$ with probability 0.8. The cross-over probability is $1/7$. If the received bit $Y=0$, the conditional probability that $'1'$ was transmitted is _____.

Open solution



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Q35. GATE ECE 2014 Set 1

Let X_1, X_2 and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $p\{X_1$ is the largest$\}$ is _____.

Open solution

Q36. GATE ECE 2014 Set 4

Parcels from sender S to receiver R pass sequentially through two post - offices. Each post - office has a probability $\frac{1}{5}$ of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post - office is _____.

Open solution

Q37. GATE ECE 2014 Set 2

Let X be a random variable which is uniformly chosen from the set of positive odd numbers less than 100. The expectation, $E[X]$, is _____.

Open solution

Q38. GATE ECE 2014 Set 3

A fair coin is tossed repeatedly till both head and tail appear at least once. The average number of tosses required is _____.

Open solution

Q39. GATE ECE 2014 Set 3

Let $X_1, X_2,$ and $X_3,$ be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $P\{X_1 + X_2 \leq X_3\}$ is _____.

Open solution

Q40. GATE ECE 2013

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

(A) $4/9$ (B) $1/2$ (C) $2/3$ (D) $5/9$

Open solution

Q41. GATE ECE 2013

Consider two identically distributed zero - mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then for all values of x

(A) $F(x) - G(x) \leq 0$ (B) $F(x) - G(x) \geq 0$

(C) $(F(x) - G(x)).x \leq 0$ (D) $(F(x) - G(x)).x \geq 0$

Open solution

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Q42. GATE ECE 2012

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

(A) $1/3$ (B) $1/2$ (C) $2/3$ (D) $3/4$

Open solution

Q43. GATE ECE 2010

A fair coin is tossed independently four times. The probability of the event "The number of times heads show up is more than the number of times tails show up" is

(A) $1/16$ (B) $1/8$ (C) $1/4$ (D) $5/16$

Open solution

Q44. GATE ECE 2009

Consider two independent random variables X and Y with identical distributions. The variables X and Y take values $0, 1$ and 2 with probability $1/2, 1/4$ and $1/4$ respectively. What is the conditional probability $P(X + Y = 2/X - Y = 0)$?

(A) 0 (B) $1/16$ (C) $1/6$ (D) 1

Open solution

Q45. GATE ECE 2009

A discrete random variable X takes value from 1 to 5 with probabilities as shown in the table. A student calculates the mean of X as 3.5 and her teacher calculates the variance to X as 1.5. Which of the following statements is true?

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K	1	2	3	4	5
P(X=K)	0.1	0.2	0.4	0.2	0.1

(A) Both the student and the teacher are right

(B) Both the student and the teacher are wrong

(C) The student is wrong but the teacher is right

(D) The student is right but the teacher is wrong

Open solution

Q46. GATE ECE 2007

An examination consists of two papers, paper 1 and paper 2. The probability of failing in paper 1 is 0.3 and that in paper 2 is 0.2. Given that a student has failed in paper 2, the probability of failing in paper 1 is 0.6. The probability of a student failing in both the papers is

(A) 0.5 (B) 0.18 (C) 0.12 (D) 0.06

Open solution

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Answer Key

Marks 1

1. **A**	2. **0.5**	3. **4**	4. **B**	5. **0.25**
6. **0.25**	7. **D**	8. **0.028**	9. **1**	10. **6**
11. **6**	12. **B**	13. **C**	14. **0.66 to 0.67**	15. **B**
16. **0.79 to 0.81**	17. **A**	18. **C**	19. **B**	20. **C**
21. **C**	22. **A**	23. **D**		

Marks 2

24. **A**	25. **7**	26. **0.585**	27. **D**	28. **0.28 to 0.32**
29. **2.5**	30. **0.33**	31. **0.07203**	32. **6**	33. **D**
34. **0.4**	35. **0.33**	36. **0.44**	37. **50**	38. **3**
39. **0.16**	40. **B**	41. **D**	42. **C**	43. **D**
44. **C**	45. **B**	46. **C**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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